

EDITOR
Dr Jerry Sanders

CONTRIBUTING EDITORS:

AUSTRALIA
Fazel Finghly

BELGIUM
Patrick van Binstinghen

CZECHOSLOVAKIA
Marian Vajnsky

POLAND
Janek Wozniak

SCANDINAVIA
John W. Peters

UK
Chris Isaac

USA
Greg Barr, Harold Hastings

CONTRIBUTORS TO THIS ISSUE

Neil Miller
Amenda Mitchell
Michael Poole

ART EDITOR

Paul Moss

ADVERTISING MANAGER

Chris Gorman

INTERNATIONAL SALES MANAGER

Isabel Van Steenckenburgh
590 H Housatonic Road, Suite 121
Thousand Oaks
CA 91320, USA
Tel (805) 499 7141
Fax (805) 499 6009

©1990 PARALLEL PUBLISHING LTD,
32 Finsbury Square,
London WC1P 5HH, UK.

EDITORIAL/ADVERTISING:

Tel +44 (0)71 387 4295
Fax +44 (0)71 388 3975

US EDITORIAL OFFICE:

Christine Gorman
1905 Blossom Hill Road,
San Jose, CA 95134
Tel (408) 263 5738
Fax (408) 263 1668

SUBSCRIPTION RATES:

One Year £29.00
BETWEEN £29.00
EUROPE (AIR MAIL) £39.00
REST OF THE WORLD (AIR MAIL) £49.00

ORIGINATOR BY

Cymbal Ltd
13 Newmarket Passage
London W1P 2HR, UK.

PRINTED BY

Jolly & Barber, Rugby, UK.

ISSN 0953 7232

PARALLEL PUBLISHING LTD IS A NEW AND EXCITING UK
PUBLICATION. READERS OF THE INTERNATIONAL JOURNAL OF
PARALLEL COMPUTING WILL BE INTERESTED TO KNOW THAT
PARALLEL PUBLISHING LTD IS A NEW AND EXCITING UK
PUBLICATION. READERS OF THE INTERNATIONAL JOURNAL OF
PARALLEL COMPUTING WILL BE INTERESTED TO KNOW THAT

TAOS: IMPOSSIBLE BUT TRUE?

Imagine an interpreted programming language where the interpretation of source code was done so quickly that it imposed no overhead on the execution of a program.

Add to that a software-based message-passing system which enabled program code and data to be moved about a multiple processor topology so quickly that, once again, the communications overhead was almost undiscernible.

Now build a parallel computer composed of heterogeneous nodes, each capable of multitasking, and set the interpretor running on each node. Then, create a compiler whose output is code for a virtual processor, about which the only known fact is that it is addressable and is running the aforementioned interpretor.

Sounds like a dream? Well, it's a group of British software engineers are to be believed, it will soon be a dream come true. This system, known as TAOS, is currently being demonstrated to sev-

eral mainstream CPU vendors, and seems to have every chance of breaking through the credibility barrier its own performance characteristics have created in the eyes and minds of many who have seen it perform at first hand. I am one of those, and I'll happily predict that TAOS is set to make a major impact on the parallel processing marketplace.

And now an administrative note. Most magazines tend to publish their April issue half way through March, their May issue half way through April, and so on. Until now, Parallelogram has resolutely insisted on publishing each issue smack in the middle of the cover month.

However, taking advantage of our reduced production timetable in the lead in to Christmas, issue 32 of Parallelogram (this one) is a combined November/December 1990 version, so that from now on we will be able to publish our issues ahead of the month on the cover.

Watch out for our January issue just before Christmas!

JS

CONTENTS VOL 2 No 32, Nov-Dec 1990.

3 EDITORIAL

4-6, 19 NEWS

Myrica goes under, Cray goes massively parallel, AMT goes to bed with Thinking Machines, Transtech goes for high performance connect, Manchester goes for Kendall Square prototype, Torque goes head-to-head with Cogent Research...

8-9 THE TERAFLOPS STAKES

As Cray Research endorses massive parallelism, Amanda Mitchell and Jerry Sanders assess some TeraFLOPS contenders.

10-11, 18 SCIENCE OR SCANDAL?

While the German press pulls the Supremum computer to pieces, other reports speak of pleasing progress. Now the European Commission must decide on how funds for the third phase of Genesis, for which Supremum GmbH is lead partner, should be spent.

12 CONFERENCE REPORT: OCCUG 13

Michael Poole brings us this assessment of the 13th meeting of the Occam User Group.

13 DIARY

14 -16 TECHNOTES: AN i860 APPLICATION

Continuing our coverage of programming for the i860, Neil Miller describes an implementation of the wave equations on multiple i860 processors.

22 - 23 WHO'S WHERE?